It is important to distinguish between problems and their solutions, or in other words, between the tasks we wish to solve and the algorithms we design to solve them. Deep learning algorithms can be applied to many problem types and tasks. Image classification and prediction tasks are common applications of deep learning because automated image processing before deep learning was very limited, given the complexity of images. But there are many other kinds of tasks we might wish to automate, such as driving a car or balancing a portfolio of stocks and other assets. Driving a car includes some amount of image processing, but more importantly the algorithm needs to learn how to *act*, not merely to classify or predict. These kinds of problems, where decisions must be made or some behavior must be enacted, are collectively called *control tasks*.

Reinforcement learning is a generic framework for representing and solving control tasks, but within this framework we are free to choose which algorithms we want to apply to a particular control task (figure 1.4). Deep learning algorithms are a natural choice as they are able to process complex data efficiently, and this is why we'll focus on *deep* reinforcement learning, but much of what you'll learn in this book is the general reinforcement framework for control tasks (see figure 1.5). Then we'll look at how you can design an appropriate deep learning model to fit the framework and solve a task. This means you will learn a lot about reinforcement learning, and you'll probably will learn some things about deep learning that you didn't know as well.

Figure 1.4. As opposed to an image classifier, a reinforcement learning algorithm dynamically interacts with data. It continually consumes data and decides what actions to take—actions that will change the subsequent data presented to it. A video game screen might be input data for an RL algorithm, which then decides which action to take using the game controller, and this causes the game to update (e.g. the player moves or fires a weapon).